UNIVERSITY of KIGALI "Unequalled Education Excellence"

CONTINUOUS ASSESSMENT TEST

GRADUATE SCHOOL

PROGRAMME: MSC-INFORMATION TECHNOLOGY

SESSION: WEEKEND

MODULE CODE AND TITLE: MIT91205: BIG DATA ANALYTICS	**LEVEL AND TRIMESTER:** LEVEL 9, TRIMESTER 3
DATE: 19/07/2026	**TIME:** 9:30AM-11:30AM **SESSION:** WEEKEND
DURATION: 2 HOURS	**TOTAL MARKS:** 100 • TO BE SCALED TO 30 MARKS
EXAMINER/ LECTURER: DR. MUSABE Jean Bosco & DR. BUGINGO EMMANUEL	**MODERATOR:** Dr Wilson Musoni

INSTRUCTIONS TO THE CANDIDATE

1. Questions must be attempted in the answer booklet provided.
2. All queries should be directed to the invigilator; do not communicate or attempt to communicate with any other candidate.
3. You have **TWO HOURS** to complete this paper. You are not allowed to leave the examination room within the **first hour** and in the last 15 minutes of this examination.
4. This is a **CLOSED BOOK** examination.
5. Answer 3 Questions; Question 1 in Section A is compulsory and choose any two (2) other questions in Section B.
6. Please, do not write on the question paper.

SECTION A (COMPULSORY QUESTION) 50 MARKS

QUESTION ONE (Read the paper below and answer the questions that follow):

DataWave Solutions is a multinational consulting firm that specializes in helping organizations transition to big data platforms. They have recently been approached by RetailMax, a traditional retail chain operating 2,000 stores across 30 countries. RetailMax generates massive amounts of data daily, including fifteen million point-of-sale transactions, five million customer loyalty interactions, five hundred thousand social media mentions, and fifty terabytes of store surveillance footage per day. Their current IT infrastructure, which relies on traditional relational databases and on-premises servers, is struggling to handle this data volume, velocity, and variety effectively. The company's IT director, Sarah Mwangi, has identified several critical challenges that are hampering the organization's ability to compete effectively in the modern retail landscape. Monthly sales reports currently take three weeks to generate, severely limiting the company's ability to respond to market trends. The customer analytics team cannot process social media data at all, meaning valuable insights about customer sentiment and brand perception are being lost. Video footage from store surveillance is deleted after seven days due to storage constraints, eliminating any possibility of long-term analysis for security or operational improvement purposes. Real-time inventory tracking is impossible with current systems, leading to stockouts and lost sales. Furthermore, data quality issues across the organization are costing the company approximately fifty million dollars annually through misallocated resources, incorrect pricing, and poor customer targeting.

Required:

a. Assume you have been appointed as the Lead Big Data Architect for RetailMax to design and implement a scalable Big Data platform capable of managing the company's rapidly growing transactional, customer, social media, inventory, and IoT data. Develop a suitable Hadoop ecosystem architecture by selecting and integrating at least six (6) Hadoop ecosystem components (e.g., HDFS, YARN, MapReduce, Hive, Pig, Spark, HBase, Kafka, Sqoop, Flume, Oozie, ZooKeeper). For each selected component, explain its practical role within the proposed architecture, justify why it is appropriate for RetailMax's operational requirements, and demonstrate how these components would interact to support efficient data ingestion, storage, processing, analytics, and decision-making.

15marks

b. RetailMax is planning to establish an enterprise-wide Data Lake that consolidates data from multiple sources, including point-of-sale transactions, e-commerce platforms, mobile applications, supplier databases, IoT sensors, customer feedback, and social media. As a Big Data Governance

Consultant, design a comprehensive Data Governance Framework for the proposed Data Lake. Your framework should include policies, standards, procedures, and technologies for managing data quality, metadata, master data, security, privacy, regulatory compliance, access control, and lifecycle management. Critically justify why each governance component is essential for ensuring trustworthy, secure, and high-quality data, and evaluate how an effective governance strategy would enable RetailMax to maximize business value, support advanced analytics, minimize operational risks, and comply with relevant data protection regulations. **(15 Marks)**

c. Critically evaluate how the Five Vs of Big Data, Volume, Velocity, Variety, Veracity, and Value, are reflected within RetailMax's data ecosystem. For each dimension, provide an appropriate quantitative or qualitative metric, explain its operational and strategic significance, and justify why each characteristic presents both opportunities and challenges for managing large-scale retail data. Furthermore, assess how effectively addressing each of the Five Vs can improve business intelligence, customer experience, operational efficiency, and organizational competitiveness. **(10 Marks)**

d. Judgmentally compare and contrast Traditional Business Intelligence (BI) systems with modern Big Data architectures in the context of RetailMax's business environment. Your discussion should identify the major architectural, technological, and analytical limitations of traditional BI systems that are constraining RetailMax's ability to process large and diverse datasets. Justify why migrating to a modern Big Data ecosystem (such as Hadoop, Spark, NoSQL databases, and cloud-based analytics platforms) would enhance scalability, real-time analytics, predictive decision-making, customer insights, and competitive advantage. Support your arguments with relevant examples from the case study. **(10 Marks)**

SECTION B (CHOOSE ONLY 2 QUESTIONS)

QUESTION TWO **25 Marks**

GlobalPay Solutions is a fintech company that processes payment transactions for e-commerce platforms across one hundred and fifty countries. They handle over one hundred million transactions daily, with peak volumes reaching ten thousand transactions per second. The system must provide strong consistency to maintain accurate account balances, high availability to ensure 99.999 percent uptime, and partition tolerance to remain resilient during network failures. These requirements represent the classic distributed systems trade-off that the CAP theorem describes. Recently, GlobalPay experienced a network partition between their data centers in Singapore and London, which had severe

consequences for the business. During this incident, some users experienced inconsistent account balances where transactions appeared to have been processed but then disappeared, while other users could not access their accounts at all due to system timeouts. The resulting damage was substantial: fifteen million dollars in customer compensation claims, a twenty-five percent drop in user trust scores as measured by post-incident surveys, regulatory investigations from three financial authorities, and approximately five million dollars in lost revenue during the outage. The CTO, Dr. James Okonkwo, has asked you to analyze this incident thoroughly and recommend architectural improvements that would prevent similar incidents in the future.

Required:

a. GlobalPay, a multinational digital payment company, recently experienced a major system outage after one of its distributed data centers became temporarily disconnected due to a network partition. During the incident, some customers were able to complete transactions while others received outdated account balances or were unable to access the payment platform. As a Distributed Systems Consultant, investigate the incident by applying the CAP Theorem. Demonstrate how the system's behavior reflects the three guarantees of Consistency, Availability, and Partition Tolerance (CAP), and explain why a distributed database cannot simultaneously guarantee all three properties during a network partition. Based on your analysis, recommend the most appropriate CAP trade-off for a global financial payment platform such as GlobalPay, providing clear technical and business justifications for your recommendation. **(10 Marks)**

b. Confer the trade-offs GlobalPay faced during the network partition, explaining whether they should have prioritized Consistency or Availability and justifying your answer with specific business considerations. Your response should consider the financial services context and weigh the relative importance of accurate balances versus continuous access for different types of stakeholders. **(8 Marks)**

c. Intricate on the consistency models available in distributed systems, including strong consistency, eventual consistency, and causal consistency, explaining which model you would recommend for GlobalPay's transaction processing and why. Your response should demonstrate an understanding of when each model is appropriate and how the choice impacts system design and user experience. **(7 Marks)**

QUESTION THREE **25MARKS**

HealthInsight is a healthcare analytics company that processes patient data from five hundred hospitals across multiple states. Their daily batch processing workload includes aggregating patient admission statistics, computing readmission rates by hospital and diagnosis, analyzing treatment outcomes across different patient populations, and generating regulatory compliance reports for various government agencies. They currently process fifty terabytes of data daily using a complex SQL-based system running on a single high-performance server, but processing takes twelve hours, which is no longer acceptable for their growing business needs. The Data Engineering Lead, Dr. Amina Hassan, has concluded that the organization needs to migrate to Hadoop's MapReduce framework to achieve the scalability and performance required for their expanding analytics workload. The migration must address the current performance bottlenecks while providing a foundation for future growth and more sophisticated analytics. Dr. Hassan has asked you to design their MapReduce jobs and optimize the batch processing pipeline to meet their evolving requirements.

Required

a. HealthInsight, a national healthcare analytics organization, is planning to implement a Hadoop-based Big Data platform to process millions of electronic health records (EHRs), laboratory results, medical images, wearable sensor data, and hospital admission records collected from hospitals across the country. As a Big Data Engineer, design a MapReduce solution for aggregating and analyzing patient data to generate regional disease statistics, identify high-risk patients, and support healthcare decision-making. Clearly demonstrate how the Mapper, Combiner, Partitioner, and Reducer phases would process the healthcare datasets, explaining the flow of intermediate data between each phase. Justify why each component is essential for improving parallel processing, reducing network traffic, balancing workload across computing nodes, and efficiently processing large-scale healthcare data within the Hadoop ecosystem. **(15 Marks)**

b. Designate the Shuffle and Sort phase in detail, explaining why this is the most expensive part of MapReduce and identifying how HealthInsight can optimize this phase for their specific workload. Your response should address the technical aspects of data movement and sorting, and provide practical optimization strategies. **(10 Marks)**

QUESTION FOUR **25MARKS**

MediaStream is a global video streaming platform with two hundred million subscribers that generates enormous data volumes daily. The platform receives one petabyte of new video content uploaded daily, generates five hundred terabytes of user watch logs, produces one hundred terabytes of user metadata, and creates fifty terabytes of recommendation algorithm output data every day. The company implemented HDFS version 3.x for their data lake, but they are now experiencing significant performance issues that are impacting their analytics capabilities and operational costs. The NameNode memory utilization is consistently at ninety-five percent capacity, threatening system stability. Small file storage is proving highly inefficient, with millions of small metadata files consuming disproportionate NameNode resources. Replication costs have reached three million dollars monthly, representing a substantial and growing operational expense. Query performance for analytics workloads is increasingly slow, frustrating the data science team and delaying business decisions. The VP of Engineering, Maria Santos, has requested a comprehensive review of their HDFS architecture and optimization strategies to address these issues while maintaining the scale and reliability that the platform requires.

Required:

a. Design the HDFS master-slave architecture, explaining the roles of NameNode, DataNode, and Secondary NameNode in MediaStream's deployment. Your response should provide a clear understanding of how these components interact and why the architecture is designed the way it is, including the specific responsibilities of each node type. **(10 Marks)**

b. Describe the impact of storing millions of small files in HDFS, analyzing why this practice is problematic for MediaStream's NameNode memory and overall system performance. Your response should address the specific technical challenges that arise from a large number of small files and provide a quantitative understanding of how this affects NameNode resource consumption. **(15 Marks)**

QUESTION FIVE **25MARKS**

RetailDirect is an e-commerce company with fifty million customers and five hundred thousand products. They have been using MapReduce for two years to process their data, but their data analysts are increasingly frustrated with writing complex Java jobs for what should be straightforward analytical queries. The Head of Analytics, Ms. Sarah Chen, has convinced the executive team to implement Hive for their data warehouse to enable SQL-like queries and accelerate the analytics workflow. Ms. Chen

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FINAL EXAMINATION

GRADUATE SCHOOL

PROGRAMME:
MSC-INFORMATION TECHNOLOGY

MODULE CODE AND TITLE: MIT91205: BIG DATA ANALYTICS	**LEVEL AND TRIMESTER:** LEVEL 9, TRIMESTER 3
DATE: 26/07/2026	**TIME:** 9:30 AM - 12:30 AM **SESSION:** WEEKEND
DURATION: 3 HOURS	**TOTAL MARKS:** 100 TO BE SCALED TO 40 MARKS
EXAMINER/ LECTURER: DR. MUSABE Jean Bosco & DR. BUGINGO EMMANUEL	**MODERATOR:** Dr Wilson Musoni

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has established clear requirements for the new system. Analysts must be able to query sales, customer, and product data easily without requiring specialized programming skills. Daily reports must run in under two hours to support time-sensitive business decisions. Historical data spanning five years must be readily accessible for trend analysis and reporting. The system must handle complex joins and aggregations efficiently as the analysis becomes increasingly sophisticated. Furthermore, user growth is twenty percent month-over-month, requiring a scalable solution that can accommodate increasing query loads. Ms. Chen has specifically requested help with performance optimization, particularly on queries that join sales data with customer demographics and product categories, which are currently the most time-consuming operations.

Required:

a. Debate the trade-offs between Hive and other query engines on Hadoop, including Pig and Spark SQL, explaining when RetailDirect should use each tool in their data processing architecture. Your response should provide practical guidance based on different workload characteristics and user requirements. **(15 Marks)**

b. Enlighten the architecture of Apache Hive, including the roles of Metastore, Driver, Compiler, and HiveServer2, and describe how RetailDirect would use each component in their data warehouse deployment. Your response should demonstrate a comprehensive understanding of how Hive transforms SQL queries into distributed processing jobs. **(10 Marks)**

ALL THE BEST!

SECTION A (COMPULSORY QUESTION) 40 MARKS

QUESTION ONE

Social Trend is a social media analytics company that provides real-time sentiment analysis and trend detection services to brands, political campaigns, and market research firms. The company processes data from multiple social media platforms, including Twitter, Facebook, Instagram, Reddit, and YouTube, ingesting approximately 500 million posts per day, which represents about 5,800 posts per second on average, with peak volumes exceeding 20,000 posts per second during major events. The company's current data processing system is based on batch processing, which means that sentiment analysis results are available with a delay of several hours. This delay is unacceptable for many of their clients, including political campaigns that need to respond to public sentiment in real-time, brands that need to manage reputation crises immediately, and financial firms that use sentiment analysis for algorithmic trading decisions. The Chief Executive Officer, Ms. Grace Mwangi, has decided to implement a real-time sentiment analysis pipeline that can provide results with a latency of less than five seconds. The pipeline must ingest streaming data from multiple social media APIs, preprocess and clean the text data, apply sentiment analysis models, detect trending topics and emerging trends, and deliver results to clients through APIs, dashboards, and alert systems. The system must handle the peak data volumes while maintaining low latency, scale horizontally as the company grows, and provide exactly-once processing guarantees to ensure data accuracy.

Required:

a. Explain the architecture of Apache Kafka, including brokers, topics, partitions, producers, and consumers, and describe how Social Trend would structure their topics for different social media platforms and data types. Your response should demonstrate a clear understanding of how to design a multi-platform Kafka architecture that handles different data formats, rates, and priorities.

(10 Marks)

b. Call the concept of consumer groups in Kafka and explain how this would enable Social Trend to process twenty thousand posts per second during peak events with sub-five-second latency. Your response should address the relationship between partitions and consumers across different platforms, how parallelism is achieved, and how consumer group rebalancing would work when processing volumes fluctuate.

(10 Marks)

c. Elaborate on the three processing guarantees available in Kafka and analyze which guarantee Social Trend should implement for their sentiment analysis pipeline, providing detailed justification based on the business context of social media analytics where accuracy and timeliness are both critical. Your response should consider the consequences of each guarantee in terms of analysis accuracy and client satisfaction.

(10 Marks)

d. Discuss the distinction between Event Time and Processing Time in Spark Structured Streaming and explain why Event Time is essential for Social Trend's sentiment analysis where social media posts may experience delays due to API rate limiting, network issues, or platform-specific delays. Your response should explain how Event Time enables accurate trend detection and sentiment analysis that reflects actual public sentiment rather than processing delays. **(10 Marks)**

20MARKS

QUESTION TWO

HealthData Systems is a healthcare technology company that provides electronic health record platforms and healthcare analytics solutions to hospitals, clinics, and healthcare networks across fifty countries. The company's platform manages data for approximately 100 million patients worldwide, including comprehensive medical histories, diagnostic images, laboratory results, genomic sequences, clinical trial data, and real-time patient monitoring from wearable devices. The company's current platform relies heavily on a monolithic relational database that is struggling to meet the growing demands of modern healthcare data management. Query performance has degraded significantly as the database has grown, with some complex clinical queries taking over thirty seconds to return results. The rigid relational schema cannot accommodate the evolving and diverse data structures required for genomic data and real-time monitoring. Scalability limitations are preventing the company from expanding into new markets, and infrastructure costs are growing at an unsustainable rate. The Chief Technology Officer, Dr. Mark Ochieng, has concluded that the company must adopt a polyglot persistence strategy using multiple NoSQL databases to address these challenges. The strategy must support the diverse data types present in healthcare while maintaining compliance with stringent healthcare data regulations such as HIPAA in the United States, GDPR in Europe, and PDPA in Asia. The solution must also ensure data security, patient privacy, and auditability.

Required:

a. Clarify the concept of polyglot persistence and analyze why HealthData Systems would benefit from using multiple database types rather than maintaining their single relational database approach. Your response should demonstrate an understanding of the strengths and weaknesses of different database technologies and explain how a polyglot approach addresses the specific challenges of healthcare data management. **(10 Marks)**

b. Compare the four types of NoSQL databases, specifically Key-Value stores, Document databases, Column-Family databases, and Graph databases, providing HealthData Systems use cases and specific technology examples for each type. Your response should address how each database type would serve different healthcare data needs, such as patient profiles, genomic data, clinical trial tracking, and medical imaging metadata. **(10 Marks)**

QUESTION THREE

InsuranceGuard is a global insurance company that processes millions of claims annually across healt auto, property, and life insurance lines. The company is facing significant challenges with fraud detection, with approximately ten percent of claims containing fraudulent elements, costing the company approximately one billion dollars annually. Current fraud detection systems use rules-based approaches and statistical models that identify individual suspicious claims but fail to detect organized fraud rings and complex fraud networks. The Chief Fraud Investigator, Mr. James Omondi, has concluded that the company needs to adopt graph-based analytics to detect fraudulent networks and patterns that are invisible to traditional approaches. Fraud rings often involve multiple individuals working together, with patterns such as shared addresses, shared phone numbers, shared medical providers, or shared repair shops. These connections can be modeled as a graph where individuals, providers, and locations are nodes, and relationships such as shared addresses or transactions are edges. The Head of Data Science, Dr. Grace Akinyi, has proposed using Spark GraphX to implement graph-based fraud detection. The solution must identify fraud rings and suspicious patterns in real-time, scale to handle the company's growing data volumes, integrate with existing fraud detection systems, and provide explainable results that can be used in legal proceedings against fraudsters.

Required:

a. Elucidate the core concepts of Spark GraphX, including VertexRDD, EdgeRDD, and Graph objects, and describe how InsuranceGuard would model their claims data as a graph for fraud detection. Your response should demonstrate a clear understanding of graph data structures and explain how nodes and edges would represent different entities and relationships in the insurance claims context.

(10 Marks)

b. Describe the PageRank algorithm in GraphX and explain how InsuranceGuard would use PageRank to identify the most influential entities in their fraud networks. Your response should explain how PageRank works, what it measures, and how high PageRank scores might indicate important entities in fraud networks such as central organizers or frequently used service providers.

(10 Marks)

QUESTION FOUR

20MARKS

GlobalHealth Analytics is a multinational healthcare data analytics company that provides insights and predictive models to healthcare providers, insurance companies, pharmaceutical companies, and government health agencies across sixty countries. The company processes data from over 2,000 hospitals, 10,000 clinics, and 50 million patients, representing one of the largest healthcare datasets globally. The company handles a diverse range of healthcare data including electronic health records

with comprehensive medical histories, clinical notes containing unstructured text, medical imaging data such as X-rays and MRIs, genomic sequencing data, real-time patient monitoring from wearable devices, and clinical trial results and drug efficacy data. This data is subject to stringent regulatory requirements including the General Data Protection Regulation in Europe, the Health Insurance Portability and Accountability Act in the United States, the Personal Data Protection Act in various Asian jurisdictions, and multiple local and national healthcare data regulations. GlobalHealth is facing serious challenges that threaten their business operations and reputation. The company experienced two significant data breaches in the past eighteen months, resulting in substantial financial losses and reputational damage. Compliance violations have resulted in fines totaling over seventy-five million dollars across multiple jurisdictions. Annual cloud infrastructure costs have exceeded thirty-five million dollars and continue to grow at approximately twenty percent annually. Data quality issues affect approximately twelve percent of clinical records, undermining the accuracy of their analytics. Audit response times average four months, creating substantial regulatory risk and delaying essential reporting to health authorities. The Chief Information Officer, Dr. Grace Akinyi, has asked you to design a comprehensive data governance and security framework that addresses all these challenges while supporting the company's continued growth and innovation in healthcare analytics.

Required:

a. Explain the key pillars of data governance and describe how GlobalHealth would implement each pillar in their healthcare data platform to ensure comprehensive data management, regulatory compliance, and data trustworthiness. Your response should address the specific challenges of healthcare data, including the need for rigorous patient privacy protections, the importance of accurate clinical data for patient safety, the requirement for comprehensive audit trails, and the need to balance data accessibility with security. **(8 Marks)**

b. Describe the six dimensions of data quality, including accuracy, completeness, consistency, timeliness, validity, and uniqueness, and explain how GlobalHealth would ensure each dimension for their diverse healthcare data sources. Your response should provide practical approaches to data quality management in the context of large-scale healthcare analytics, including specific techniques for validating clinical data, detecting data anomalies, and maintaining data integrity across multiple systems. **(12 Marks)**

QUESTION FIVE

CloudTech Solutions is a cloud consulting firm that helps organizations optimize their cloud infrastructure costs. The company has observed that many organizations are overspending significantly on their cloud big data platforms, with costs often exceeding budgets by fifty percent or more. These

cost overruns are driven by several factors including inefficient storage utilization, over-provisi compute resources, lack of cost visibility, and suboptimal query patterns. The company has be engaged by several clients to help them optimize their cloud big data costs. These clients use various cloud platforms including AWS, Google Cloud, and Azure, and they run a variety of big data workloads including data lakes, data warehouses, real-time streaming, and machine learning. Each client has unique requirements but shares the common goal of reducing costs while maintaining or improving performance. The Chief Cloud Architect, Ms. Grace Ochieng, has asked you to develop a comprehensive cost optimization framework for cloud big data platforms that addresses the various cost drivers and provides practical strategies for reducing costs. The framework must be applicable across different cloud platforms and workload types, and it must balance cost reduction with performance and reliability requirements.

Required:

a. Clarify the key cost drivers in cloud big data platforms, including storage costs, compute costs, and data transfer costs, and analyze how each driver contributes to overall cloud spending. Your response should address the specific characteristics of big data workloads that make them expensive to run in the cloud.

(10 Marks)

b. Describe storage optimization strategies for cloud big data platforms, including storage tiering, data compression, data lifecycle management, and data deduplication, and explain how these strategies reduce storage costs. Your response should address how different data types and access patterns should be managed to optimize storage costs.

(10 Marks)

QUESTION SIX

20MARKS

FinSecure is a multinational financial services corporation that processes millions of banking and investment transactions daily across retail banking, investment banking, and wealth management divisions. The company is currently facing significant challenges in fraud detection, customer churn prevention, and risk assessment due to the limitations of their traditional analytics platforms.

Fraud detection currently uses rules-based systems that are easily circumvented by sophisticated criminals, with approximately fifteen percent of fraudulent transactions going undetected. Customer churn prediction models rely on outdated methods that identify at-risk customers only after they have already initiated the process of leaving the bank. Risk assessment models take over 48 hours to generate credit scores, limiting the bank's ability to approve loans quickly in competitive markets. These challenges are costing the company substantial revenue and putting them at a competitive disadvantage.

The Chief Data Scientist, Dr. Sarah Kilonzo, has decided to implement a comprehensive machine learning platform using Spark MLlib for scalable model training and deployment. The platform must provide sub-second scoring latency for transaction fraud detection, support daily model retraining to adapt to evolving fraud patterns, enable real-time customer churn prediction, and scale to accommodate the company's growing data volumes.

Required:

a. Describe the feature engineering pipeline in Spark MLlib, including the roles of Vector Assembler, StandardScaler, and StringIndexer, and explain how these components would be applied to FinSecure's transaction data to prepare it for machine learning modeling. Your response should address the specific challenges of financial data, including handling categorical variables such as merchant categories and transaction types, normalizing numerical features such as transaction amounts and account balances, and dealing with missing or incomplete data. **(10 Marks)**

b. Discuss the importance of feature engineering in machine learning and explain how FinSecure would use Spark MLlib's feature transformers to handle categorical variables, normalize numerical features, and combine features for their fraud detection models. Your response should address the specific challenges of financial data and explain how proper feature engineering improves model performance. **(10 Marks)**

END

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FINAL EXAMINATION

GRADUATE SCHOOL

PROGRAMME: MSC-INFORMATION TECHNOLOGY
SESSION: WEEKEND

MODULE CODE AND TITLE: MSCIT 9128: MACHINE LEARNING AND BIG DATA ANALYTICS	**LEVEL AND TRIMESTER:** LEVEL 9, TRIMESTER 2
DATE: 26/10/2025	**TIME:** 9:30 AM - 12:30 PM **SESSION:** WEEKEND
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SECTION A (COMPULSORY QUESTION) 40 MARKS

QUESTION ONE

In an effort to promote sustainable food production, a regional agricultural board has launched a Big Data–driven precision farming initiative. The project seeks to optimize resource use and enhance crop yield prediction across large-scale farms using integrated IoT, remote sensing, and analytics technologies. Sensors deployed throughout the fields continuously collect data on soil nutrients, temperature, moisture levels, and pest activity, while drones and satellites provide high-resolution imagery of crop health.

All incoming data is stored and processed through distributed systems such as Hadoop Distributed File System (HDFS) and cloud-based platforms like Amazon S3 to handle the massive volume, velocity, and variety of agricultural data. To enable near real-time insights, Apache Spark supports in-memory analytics, while Apache Kafka manages live data streams from IoT devices.

The analytical phase involves predictive models to forecast yield and disease outbreaks, while prescriptive analytics guides optimal irrigation schedules and fertilizer application. By combining machine learning with spatial and temporal data, the project aims to create an intelligent decision support system that assists farmers and policymakers alike. Nevertheless, the team must address data quality challenges arising from faulty sensors, inconsistent satellite feeds, and varying environmental conditions. Ensuring veracity and reliability of agricultural data is crucial for accurate decision-making and long-term sustainability.

Required:

1. Identify the 5Vs of Big Data present in this project. **(5 Marks)**
2. Describe how data sources, storage, processing, and analysis can be integrated into the city's Big Data architecture. **(6 Marks)**
3. Compare the suitability of Apache Hadoop and Apache Spark for this project. **(6 Marks)**
4. Illustrate how Kafka could be implemented to manage real-time traffic updates. **(5 Marks)**
5. Analyze the role of predictive and prescriptive analytics in improving city operations. **(6 Marks)**
6. Evaluate the potential risks of data veracity and recommend methods to enhance data quality. **(6 Marks)**
7. Propose one innovative application of spatial analytics in this smart city project. **(6 Marks)**

x dynamic curb space

x time stamp
x drive malfunction
x

SECTION B (SELECT ONLY THREE) 60 MARKS

QUESTION TWO

A local bank wants to detect unusual transactions that could indicate potential fraud. The bank processes thousands of transactions daily, but fraudulent ones are rare. Machine learning is being used to analyze transaction patterns and raise alerts for suspicious activities.

Required:

1. Explain why stream processing might be preferred over batch processing. **(4 marks)**
2. Classify this task as classification or regression and justify. **(4 marks)**
3. Identify two evaluation metrics suitable for imbalanced data. **(4 marks)**
4. Discuss how real-time alerts could reduce losses. **(4 marks)**
5. Suggest one strategy to reduce false alarms in fraud detection. **(4 marks)**

QUESTION THREE

An online learning platform collects student engagement data such as login frequency, video completion rate, and quiz performance. The institution wants to use analytics to predict dropout risk and recommend learning support.

Required:

1. Describe how predictive analytics could identify at-risk students. **(4 marks)**
2. Compare descriptive and predictive analytics in this context. **(4 marks)**
3. Suggest one data visualization tool for tracking learning progress. **(4 marks)**
4. Explain how ethical considerations apply when analyzing student data. **(4 marks)**
5. Propose ways to balance model accuracy and fairness. **(4 marks)**

QUESTION FOUR

A hospital uses machine learning models to analyze X-ray images for early disease detection. They plan to combine structured data (age, gender, symptoms) with image data to enhance diagnostic accuracy.

Required:

1. Explain how deep learning can assist in image recognition tasks. **(4 marks)**
2. Describe how data preprocessing improves image quality before modeling. **(4 marks)**

reliable

3. Identify challenges of model interpretability in healthcare. **(4 marks)**

4. Discuss how bias might appear in medical datasets. **(4 marks)**

5. Suggest techniques to ensure fairness and transparency. **(4 marks)**

QUESTION FIVE

A retail company wants to optimize its inventory and predict demand across multiple branches. It plans to use sales data, weather conditions, and promotional events for analysis.

Required:

1. Explain how temporal analytics could improve demand forecasting. **(4 marks)**

2. Describe how spatial analytics can help identify regional demand patterns. **(4 marks)**

3. Differentiate between operational and descriptive analytics in this case. **(4 marks)**

4. Propose how real-time dashboards could support decision-making. **(4 marks)**

5. Evaluate the limitations of relying solely on analytics for strategic decisions. **(4 marks)**

QUESTION SIX

A local bank is on a mission to enhance its security measures by detecting unusual transactions that might signal potential fraud. With thousands of transactions processed each day, the bank faces the challenge of identifying rare fraudulent activities amidst an ocean of legitimate ones. To tackle this issue, the bank is employing machine learning techniques to meticulously analyze transaction patterns. This advanced technology enables the bank to raise alerts for any activities that appear suspicious, allowing for quicker response and prevention of potential financial crime.

Required:

1. Explain how sentiment analysis extracts opinions from social media data. **(4 marks)**

2. Discuss the benefits of combining text analytics and visualization tools. **(4 marks)**

3. Suggest how real-time analytics can enhance marketing responses. **(4 marks)**

4. Assess challenges posed by fake accounts and misinformation. **(4 marks)**

5. Recommend measures to improve data reliability. **(4 marks)**

END